A. Program File

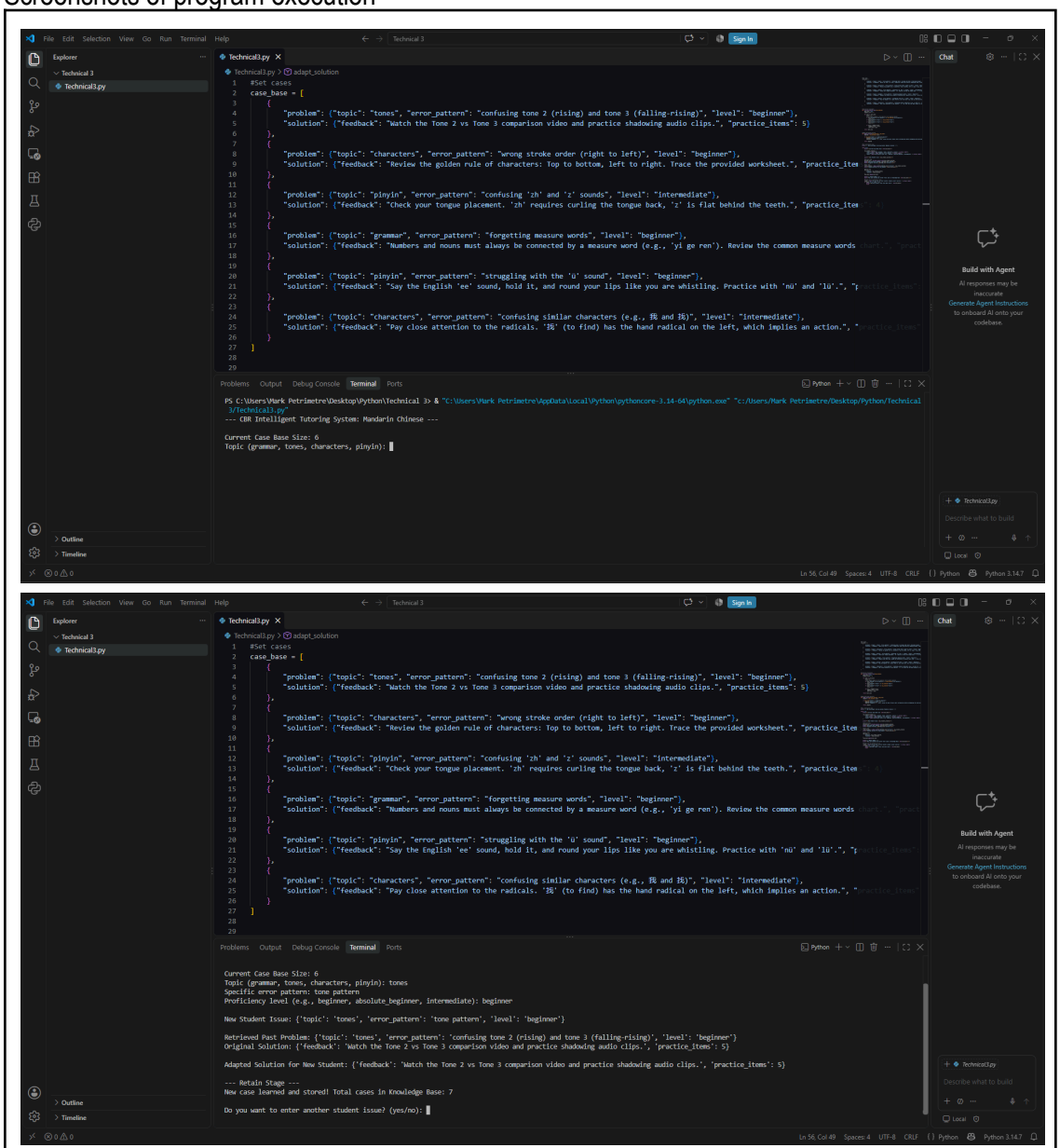
- Python

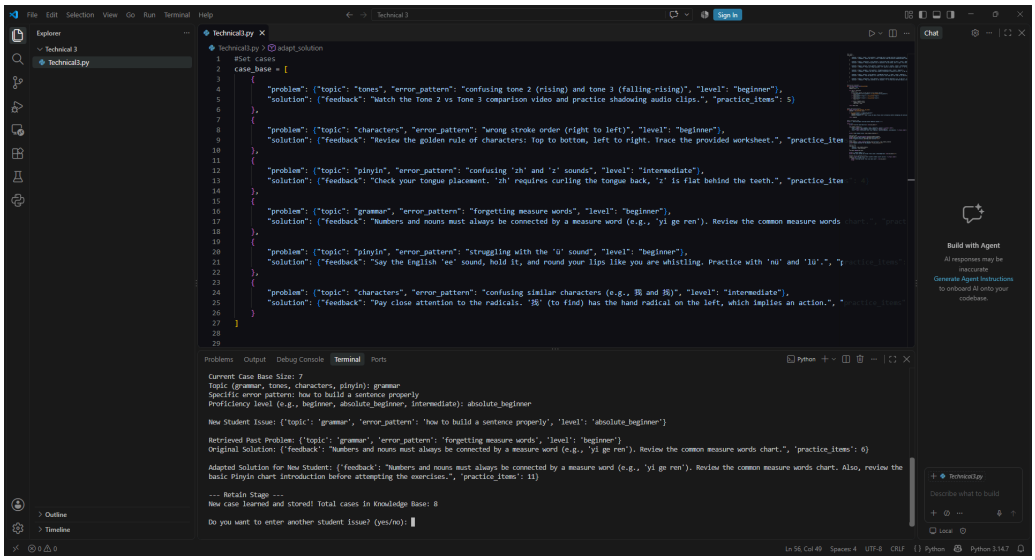
B. Short Documentation (1–2 pages)

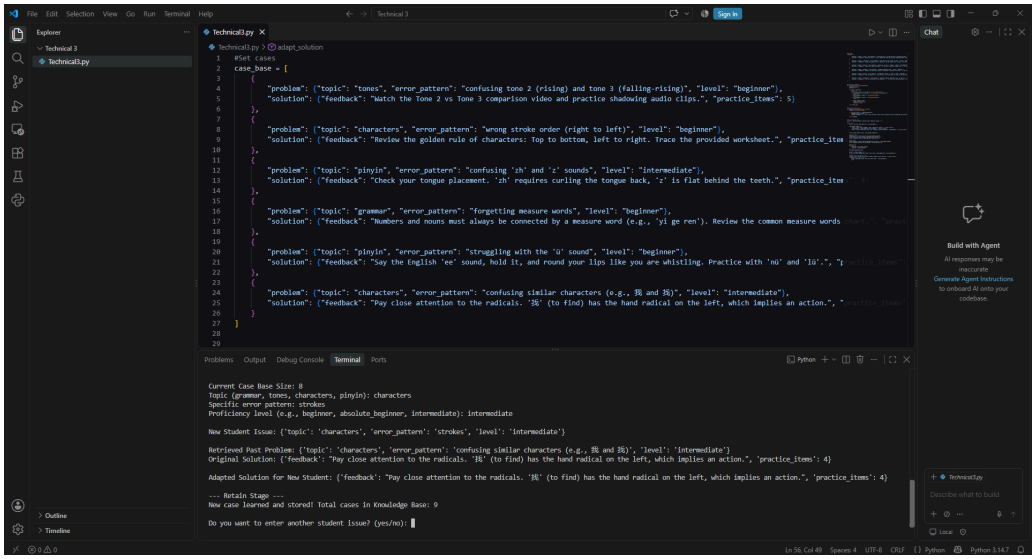
Include:

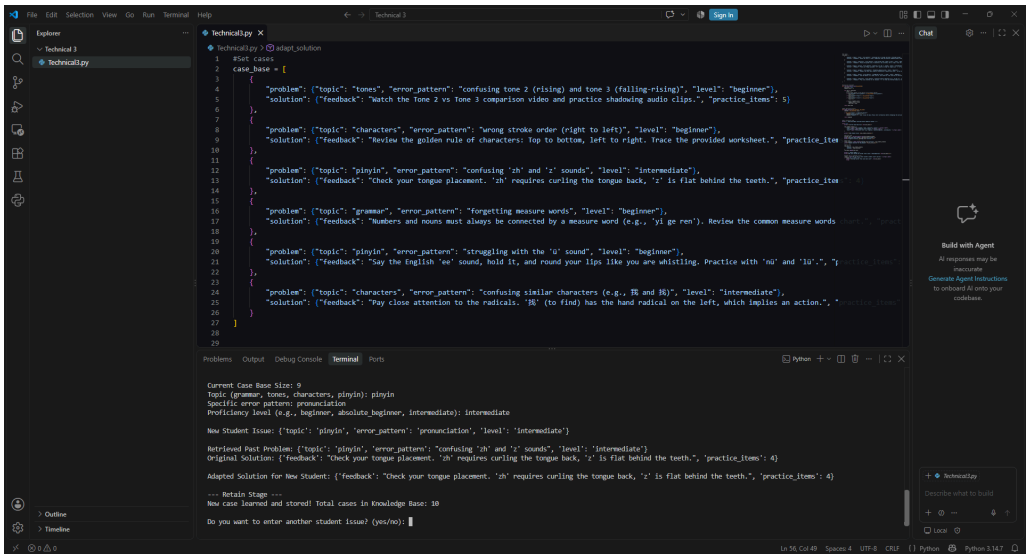
- Title of chosen case study
  - Intelligent Tutoring System (ITS) - Mandarin Chinese
- Description of your case base
  - Our knowledge base uses a list of dictionaries to store 6 past cases of students learning Mandarin Chinese. Each case records a specific student problem, categorized by the topic (tones, characters, pinyin, or grammar), the exact error_pattern (such as confusing the 'zh' and 'z' sounds, or forgetting measure words), and the student's proficiency level. The paired solutions contain targeted instructional feedback and a recommended number of practice_items to resolve the specific struggle.
- Explanation of similarity method
  - The similarity assessment is handled by the retrieve_similar_case function, which iterates through the case base and calculates a match score for each past case. The system prioritizes the most specific issue by awarding the highest weight (3 points) for an exact match on the error_pattern. It also awards 2 points for a matching topic and 1 point for a matching proficiency level. To prevent the scoring logic from breaking due to user typos or extra spaces, the system sanitizes all user inputs using the .strip() and .lower() methods before running the comparison. The case with the highest total score is retrieved as the best match.
- Adaptation logic
  - The program adapts the retrieved solution by checking the new student's proficiency level against the original case. If the user inputs their level as "absolute_beginner", the system assumes they need more repetition and foundational support than a standard beginner. It modifies the old solution by mathematically adding 5 extra practice items to their workload and appending a new sentence to the feedback string, specifically advising them to review the basic Pinyin chart before starting the exercises.

- Screenshots of program execution




Technical3.py
#Set cases
case_base = [
{
"problem": {"topic": "tones", "error_pattern": "confusing tone 2 (rising) and tone 3 (falling-rising)", "level": "beginner"},
"solution": {"feedback": "Watch the Tone 2 vs Tone 3 comparison video and practice shadowing audio clips.", "practice_items": 5}
},
{
"problem": {"topic": "characters", "error_pattern": "wrong stroke order (right to left)", "level": "beginner"},
"solution": {"feedback": "Review the golden rule of characters: Top to bottom, left to right. Trace the provided worksheet.", "practice_item
},
{
"problem": {"topic": "pinyin", "error_pattern": "confusing 'zh' and 'z' sounds", "level": "intermediate"},
"solution": {"feedback": "Check your tongue placement. 'zh' requires curling the tongue back, 'z' is flat behind the teeth.", "practice_item
},
{
"problem": {"topic": "grammar", "error_pattern": "forgetting measure words", "level": "beginner"},
"solution": {"feedback": "Numbers and nouns must always be connected by a measure word (e.g., 'yi ge ren'). Review the common measure words
},
{
"problem": {"topic": "pinyin", "error_pattern": "struggling with the 'ü' sound", "level": "beginner"},
"solution": {"feedback": "Say the English 'ee' sound, hold it, and round your lips like you are whistling. Practice with 'nü' and 'lü'.", "
},
{
"problem": {"topic": "characters", "error_pattern": "confusing similar characters (e.g., 我 and 找)", "level": "intermediate"},
"solution": {"feedback": "Pay close attention to the radicals. '找' (to find) has the hand radical on the left, which implies an action.", "
}
]
Problems Output Debug Console Terminal Ports
Current Case Base Size: 7
Topic (grammar, tones, characters, pinyin): grammar
Specific error pattern: how to build a sentence properly
Proficiency level (e.g., beginner, absolute_beginner, intermediate): absolute_beginner
New Student Issue: {'topic': 'grammar', 'error_pattern': 'how to build a sentence properly', 'level': 'absolute_beginner'}
Retrieved Past Problem: {'topic': 'grammar', 'error_pattern': 'forgetting measure words', 'level': 'beginner'}
Original Solution: {'feedback': "Numbers and nouns must always be connected by a measure word (e.g., 'yi ge ren'). Review the common measure words chart.", 'practice_items': 6}
Adapted Solution for New Student: {'feedback': "Numbers and nouns must always be connected by a measure word (e.g., 'yi ge ren'). Review the common measure words chart. Also, review the basic Pinyin chart introduction before attempting the exercises.", 'practice_items': 11}
--- Retain Stage ---
New case learned and stored! Total cases in Knowledge Base: 8
Do you want to enter another student issue? (yes/no):
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.


Technical3.py
#Set cases
case_base = [
{
"problem": {"topic": "tones", "error_pattern": "confusing tone 2 (rising) and tone 3 (falling-rising)", "level": "beginner"},
"solution": {"feedback": "Watch the Tone 2 vs Tone 3 comparison video and practice shadowing audio clips.", "practice_items": 5}
},
{
"problem": {"topic": "characters", "error_pattern": "wrong stroke order (right to left)", "level": "beginner"},
"solution": {"feedback": "Review the golden rule of characters: Top to bottom, left to right. Trace the provided worksheet.", "practice_item
},
{
"problem": {"topic": "pinyin", "error_pattern": "confusing 'zh' and 'z' sounds", "level": "intermediate"},
"solution": {"feedback": "Check your tongue placement. 'zh' requires curling the tongue back, 'z' is flat behind the teeth.", "practice_item
},
{
"problem": {"topic": "grammar", "error_pattern": "forgetting measure words", "level": "beginner"},
"solution": {"feedback": "Numbers and nouns must always be connected by a measure word (e.g., 'yi ge ren'). Review the common measure words
},
{
"problem": {"topic": "pinyin", "error_pattern": "struggling with the 'ü' sound", "level": "beginner"},
"solution": {"feedback": "Say the English 'ee' sound, hold it, and round your lips like you are whistling. Practice with 'nü' and 'lü'.", "
},
{
"problem": {"topic": "characters", "error_pattern": "confusing similar characters (e.g., 我 and 找)", "level": "intermediate"},
"solution": {"feedback": "Pay close attention to the radicals. '找' (to find) has the hand radical on the left, which implies an action.", "
}
]
Problems Output Debug Console Terminal Ports
Current Case Base Size: 8
Topic (grammar, tones, characters, pinyin): characters
Specific error pattern: strokes
Proficiency level (e.g., beginner, absolute_beginner, intermediate): intermediate
New Student Issue: {'topic': 'characters', 'error_pattern': 'strokes', 'level': 'intermediate'}
Retrieved Past Problem: {'topic': 'characters', 'error_pattern': 'confusing similar characters (e.g., 我 and 找)', 'level': 'intermediate'}
Original Solution: {'feedback': "Pay close attention to the radicals. '找' (to find) has the hand radical on the left, which implies an action.", 'practice_items': 4}
Adapted Solution for New Student: {'feedback': "Pay close attention to the radicals. '找' (to find) has the hand radical on the left, which implies an action.", 'practice_items': 4}
--- Retain Stage ---
New case learned and stored! Total cases in Knowledge Base: 9
Do you want to enter another student issue? (yes/no):
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.


Technical3.py
#Set cases
case_base = [
{
"problem": {"topic": "tones", "error_pattern": "confusing tone 2 (rising) and tone 3 (falling-rising)", "level": "beginner"},
"solution": {"feedback": "Watch the Tone 2 vs Tone 3 comparison video and practice shadowing audio clips.", "practice_items": 5}
},
{
"problem": {"topic": "characters", "error_pattern": "wrong stroke order (right to left)", "level": "beginner"},
"solution": {"feedback": "Review the golden rule of characters: Top to bottom, left to right. Trace the provided worksheet.", "practice_item
},
{
"problem": {"topic": "pinyin", "error_pattern": "confusing 'zh' and 'z' sounds", "level": "intermediate"},
"solution": {"feedback": "Check your tongue placement. 'zh' requires curling the tongue back, 'z' is flat behind the teeth.", "practice_item
},
{
"problem": {"topic": "grammar", "error_pattern": "forgetting measure words", "level": "beginner"},
"solution": {"feedback": "Numbers and nouns must always be connected by a measure word (e.g., 'yi ge ren'). Review the common measure words
},
{
"problem": {"topic": "pinyin", "error_pattern": "struggling with the 'ü' sound", "level": "beginner"},
"solution": {"feedback": "Say the English 'ee' sound, hold it, and round your lips like you are whistling. Practice with 'nü' and 'lü'.", "
},
{
"problem": {"topic": "characters", "error_pattern": "confusing similar characters (e.g., 我 and 找)", "level": "intermediate"},
"solution": {"feedback": "Pay close attention to the radicals. '找' (to find) has the hand radical on the left, which implies an action.", "
}
]
Problems Output Debug Console Terminal Ports
Current Case Base Size: 9
Topic (grammar, tones, characters, pinyin): pinyin
Specific error pattern: pronunciation
Proficiency level (e.g., beginner, absolute_beginner, intermediate): intermediate
New Student Issue: {'topic': 'pinyin', 'error_pattern': 'pronunciation', 'level': 'intermediate'}
Retrieved Past Problem: {'topic': 'pinyin', 'error_pattern': "confusing 'zh' and 'z' sounds", 'level': 'intermediate'}
Original Solution: {'feedback': "Check your tongue placement. 'zh' requires curling the tongue back, 'z' is flat behind the teeth.", 'practice_items': 4}
Adapted Solution for New Student: {'feedback': "Check your tongue placement. 'zh' requires curling the tongue back, 'z' is flat behind the teeth.", 'practice_items': 4}
--- Retain Stage ---
New case learned and stored! Total cases in Knowledge Base: 10
Do you want to enter another student issue? (yes/no):
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.

C. Pair Reflection (5–7 sentences)
Discuss:

- What each partner contributed
- What challenges you encountered
- What you learned about CBR

- Formoso started out by setting up our base code and getting the initial logic down, while Petrimetre handled running the scripts, testing the inputs, and giving feedback on how to make it work better. The main challenge we ran into was balancing the scoring weights for the similarity check and handling edge cases where a beginner case needed extra adaptation. It was also tricky figuring out how to loop the prompt properly so new cases would actually get appended and saved to the case base without resetting each time. Through this, we learned that Case-Based Reasoning is basically about solving new problems by pulling from past experiences instead of just hardcoding static if-else rules. We also realized how important the retain step is, since the system genuinely gets smarter the more student issues it processes and saves.